Reference Material on Mathematics-I (MTH1001)

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Created By: - Department of Mathematics

Department of Mathematics

C. V. Raman Global University, Odisha

Mathematics-I (MTH1001)

Unit-II

Functions of two and more Variables and Special Functions:

Functions of two or more several variables: limit, continuity and differentiability, homogenous functions and Euler's theorem, higher order partial derivatives and Taylor's series, maximum and minimum values, beta, gamma functions and error functions**.**

TEXT BOOK:

Differential Calculus by Shanti Narayan and P.K. Mittal

1. Limit and Continuity

Introduction. The statement that a function of x is defined in a certain interval means that to each value of x belonging to the interval there corresponds a value of the function. *The value of the function for any particular value of x may be quite independent of the value of the function for another value of x* and no relationship in the various values of a function is implied in the definition of a function as such.

Thus, if x_1, x_2 are any two values of the independent variable so that $f(x_1)$, $f(x_2)$ are the corresponding values of the function $f(x)$, then $|f(x_2)-f(x_1)|$ may be large even though $|x_2-x_1|$ is small. It is because the value $f(x_2)$ assigned to the function for $x=x_2$ is quite independent of the value $f(x_1)$ of the function for $x=x_1$.

We now propose to study the change $|f(x_2)-f(x_1)|$ relative to the change $|x_2-x_1|$ by introducing the notion of continuity and discontinuity of a function.

In the next article, we first analyse the intuitive notion of *continuity* that we already possess and then state its precise analytical meaning in the form of a definition.

2. Continuity of a function:

The function $f(x)$ will vary continuously at $x=c$ if, as x changes from c to either side of it, the change in the value of the function is not sudden, *i.e.*, the change in the value of the function is *small* if only the change in the value of x is also *small*.

We consider a value $c+h$ of x belonging to the interval $[a, b]$. Here, h, which is the change n the independent variable x, may be positive or negative.

Then, $f(c+h)-f(c)$, is the corresponding change in the dependent variable $f(x)$, which, again, may be positive or negative.

For continuity, we require that $c+h)-f(c)$ should be numerically small, if h is numerically small. This means that $|f(c+h)-f(c)|$ can be made as small as we like by taking $|h|$ sufficiently small.

A function $f(x)$ is continuous at $x=c$ if, corresponding to any positive number, ϵ, arbitrarily assigned, there exists a positive number δ such that

$$|f(c+h)-f(c)|<\epsilon$$

for all values of, h, such that

$$|h|<\delta.$$

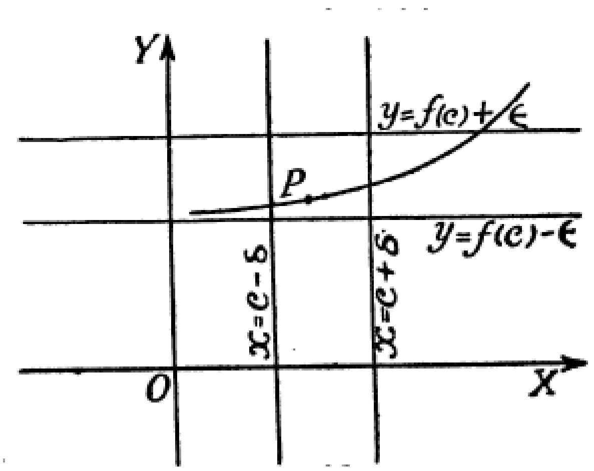
This means that $f(c+h)$ lies between $f(c)-\epsilon$ and $f(c)+\epsilon$ for all values of h lying between $-\delta$ and δ.

Geometrical Meaning of Continuity:

Draw a graph of a function the function f(x). Consider the point P[c, f(c)] on it.

Draw the line $y=f(c)-\epsilon$, $y=f(c)+\epsilon$ -------(1)

which is lie on different sides of the point P and parallel to x-axis.



The continuity of f(x) for x=c, required to draw two lines

$$x=c-\delta,\ x=c+\delta \quad \text{-----(2)}$$

which are parallel to y-axis and which lie around the line $x=c$, such that every point of the graph between the two lines (2) lies also between the two lines (1).

Continuity of a function in an interval:

A function f(x) is continuous in an interval [a, b], if it is continuous at every point thereof.

Discontinuity:

A function f(x) which is not continuous for x=c is said be discontinuous for x=c.

Examples

1. Show that f(x)=3x+1 is continuous at x=1.

Solution:

Here,

$$f(1)=,4$$

and

$$f(x)-f(1)=3x+1-4=3(x-1).$$

We will now attempt to make the numerical value of this difference smaller than any preassigned positive number, *say* ·001.

(*i*) Let $x>1$, so that $3(x-1)$ is positive and is, therefore, the numerical value of $f(x)-f(1)$.

Now

$$|f(x)-f(1)|=3(x-1)<\cdot 001,$$

if

$$x-1<\cdot 001/3$$

i e., if

$$x<1+\cdot 001/3=1\cdot 0003. \qquad \ldots(i)$$

(*ii*) Let $x<1$, so that $3(x-1)$ is negative and the numerical value of $f(x)-f(1)$ is $3(1-x)$.

Now

$$|f(x)-f(1)|=3(1-x)<\cdot 001,$$

if

$$1-x<\cdot 001/3,$$

i.e., if

$$1-\cdot 001/3<x,$$

or

$$1-\cdot 000\dot{3}<x.$$

Combining (*i*) and (*ii*), we see that

$$|f(x)-f(1)|<\cdot 001,$$

for all values of x such that

$$1-\cdot 000\dot{3}<x<1+\cdot 000\dot{3}.$$

The test of continuity for $x=1$ is thus satisfied for the particular value $\cdot 001$ of ε. We may similarly show that the test is true for the other particular values of ε also.

The *complete* argument is, however, as follows :—

Let ε be *any* positive number. We have

$$|f(x)-f(1)|=3|x-1|.$$

Now

$$3|x-1|<\varepsilon$$

if

$$|x-1|<\varepsilon/3,$$

i.e, if

$$1-\varepsilon/3<x<1+\varepsilon/3.$$

Thus, there exists an interval $(1-\varepsilon/3,\ 1+\varepsilon/3)$ around 1 such that for every value of x in this interval, the numerical value of the difference between $f(x)$ and $f(1)$ is less than a preassigned positive number ε. Here $\delta=\varepsilon/3$.

Hence $f(x)$ is continuous for $x=1$.

2. *Show that $f(x)=3x^2+2x-1$ is continuous for $x=2$.*

Solution:

Let ε be any *given* positive number. We have

$$f(2)=15.$$

$$\therefore \quad |f(x)-f(2)| = |3x^2+2x-1-15|$$
$$= |x-2|\ |3x+8|.$$

We suppose that x lies between 1 and 3. For values of x between 1 and 3, $3x+8$ is positive and less than $(3\cdot3+8)=17$.

Thus when

$$1<x<3,$$

we have

$$|f(x)-f(2)|<17\,|x-2|.$$

Now

$$17\,|x-2|<\varepsilon,$$

if

$$|x-2|<\varepsilon/17.$$

Thus, we see that there exists a positive number $\varepsilon/17$ such that

$$|f(x)-f(2)|<\varepsilon$$

when

$$|x-2|<\varepsilon/17.$$

Therefore $f(x)$ is continuous for $x=2$.

3. *Prove that sin x is continuous for every value of x.*

Solution:

Let ε be any *arbitrarily assigned positive number.* We have

$$|f(x)-f(c)| = |\sin x - \sin c|$$
$$= \left|2 \cos \frac{x+c}{2} \sin \frac{x-c}{2}\right|$$
$$= \left|2 \cos \frac{x+c}{2}\right| \left|\sin \frac{x-c}{2}\right|$$

Now $\left|\cos \frac{x+c}{2}\right| \leqslant 1$ for every value of x and c.

Also $$\left|\sin \frac{x-c}{2}\right| \leqslant \left|\frac{x-c}{2}\right|.$$

(From Elementary Trigonometry)

Thus we have

$$\left|\sin x - \sin c\right| = 2 \left|\cos \frac{x+c}{2}\right| \left|\sin \frac{x-c}{2}\right|$$
$$\leqslant 2.1. \left|\frac{x-c}{2}\right| = \left|x-c\right|.$$

Hence $|\sin x - \sin c| < \varepsilon$ when $|x-c| < \varepsilon$.

Thus, there exists an interval $(c-\varepsilon, c+\varepsilon)$ around c such that for every value of x in this interval

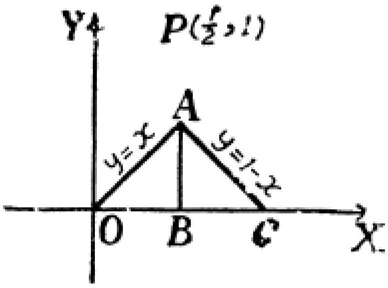
$$|\sin x - \sin c| < \varepsilon.$$

Hence, $\sin x$ is continuous for $x=c$ and, therefore, also for every value of x ; c being any number.

4. *Show that the function $f(x)$, as defined below, is discontinuous at $x=\frac{1}{2}$.*

$$f(x)=\begin{cases} x, & \text{when } 0\leqslant x<\frac{1}{2}\ ; \\ 1, & \text{when } x=\frac{1}{2}\ ; \\ 1-x, & \text{when } \frac{1}{2}<x<1. \end{cases}$$

Solution:



The graph consists of the point P ($\frac{1}{2}$, 1) and the lines OA, AC excluding the point A. Our intuition immediately suggests that there is a discontinuity at A ; there being a gap in the graph at A.

Also, analytically, we can see that for every value of x, round about $x=\frac{1}{2}$, $f(x)$ differs from $f(\frac{1}{2})$ which is equal to 1, by a number greater than $\frac{1}{2}$ so that there is no question of making the difference between $f(x)$ and $f(\frac{1}{2})$ less than *any* positive number arbitrarily assigned.

Therefore $f(x)$ is discontinuous at $x=\frac{1}{2}$.

3. Limit:

A function $f(x)$ is said to tend to a limit, l, as x tends to c, if, corresponding to any positive number ε, arbitrarily assigned, there exists a positive number; δ, such that for every value of x in the interval $(c-\delta, c+\delta)$, other than c, $f(x)$ differs from l numerically by a number which is less than ε, i.e.,

$$|f(x)-l| < \varepsilon,$$

for every value of x, other than c, such that

$$|x-c| < \delta.$$

Left hand limit and Right hand limit:

$$\lim_{x \to (c+0)} f(x) = l. \qquad \lim_{x \to (c-0)} f(x) = l$$

A function $f(x)$ is said to tend to a limit, l, as x tends to, c, **from above,** *if, corresponding to any positive number, ε, arbitrarily assigned, there exists a positive number, δ, such that*

$$|f(x)-l| < \varepsilon$$

whenever

$$c < x < c+\delta.$$

In this case, we write

$$\lim_{x \to (c+0)} f(x) = l,$$

and say that, l is the right handed limit of $f(x)$.

A function $f(x)$ is said to tend to a limit, l, as x tends to c **from below,** *if corresponding to any positive number, ε, arbitrarily assigned, their exists a positive number, δ, such that*

$$|f(x)-l| < \varepsilon,$$

whenever

$$c-\delta < x < c.$$

In this case, we write

$$\lim_{x \to (c-0)} f(x) = l$$

and say that, l, is the left handed limit of $f(x)$.

From above it at once follows that

$$\lim_{x \to c} f(x) = l,$$

if, and only if,

$$\lim_{x \to (c+0)} f(x) = l = \lim_{x \to (c-0)} f(x)$$

Examples:

Example-1:

Examine the limit of the function

$$\frac{x^2-1}{x-1},$$

as x tends to 1.

The function is defined for every value of x

$$y = \frac{x^2-1}{x-1} = x+1, \text{ when } x \neq 1.$$

Solution:

Case 1. Firstly consider the behaviour of the values of y *for values of x greater than 1.*

Clearly, the variable y is greater than 2 when x is greater than 1.

If x, while remaining greater than 1, takes up values whose difference from 1 constantly diminishes, then y, while remaining greater than 2, takes up values whose difference from 2 constantly diminishes also.

In fact, difference between y and 2 can be made as small as we like by taking x sufficiently near 1.

For instance, consider the number ·001.

Then

$$|y-2| = y-2 = x+1-2 < \cdot 001.$$

if $$x < 1 \cdot 001.$$

Thus, for every value of x which is greater than 1 and less than 1·001, the absolute value of the difference between y and 2 is less than the number ·001 which we had arbitrarily selected.

Instead of the particular number ·001, we now consider any positive number ε.

Then

$$y-2=x-1<\varepsilon,$$

if

$$x<1+\varepsilon.$$

Thus, there exists an interval $(1, 1+\varepsilon)$, *such that the value of* y, *for any value of* x *in this interval, differs from 2 numerically, by a number which is smaller than the positive number* ε, *selected arbitrarily.*

Thus the limit of y *as* x *approaches 1 through values greater than 1, is 2 and we have*

$$\lim_{x\to(1+0)} y=2.$$

Case II. We now consider the behaviour of the values of y *for values of* x *less than* 1.

When x is less than 1, y is less than 2.

If, x, while remaining less than 1, takes up values whose difference from 1 constantly diminishes, then y, while remaining less than 2, takes up values whose difference from 2 constantly diminishes also.

Let, now, ε be any arbitrarily assigned positive number, however small.

We then have

$$|y-2|=2-y=2-(x+1)=1-x<\varepsilon,$$

if

$$1-\varepsilon<x,$$

so that for every value of x less than 1 but $>1-\varepsilon$, the absolute value of the difference between y and 2 is less than the number ε.

Thus there exists *an interval* $(1-\varepsilon, 1)$ *such that the value of* y, *for any* x *in the interval, differs from 2 numerically by a number which is smaller than the arbitrarily selected positive number* ε.

Thus the *limit of* y, *as* x *approaches* 1, *through values less than* 1, *is* 2 and we write

$$\lim_{x\to(1-0)} y=2.$$

Case III. Combining the conclusions arrived at in the last two cases, we see that corresponding to any arbitrarily assigned positive number ε, there exists an interval $(1-\varepsilon,\ 1+\varepsilon)$ around 1, such that for every value of x in this interval, *other than* 1, y differs from 2 numerically by a number which is less than ε, *i.e.*, we have

$$|y-2|<\varepsilon$$

or any x, other than 1, such that

$$|x-1|<\varepsilon.$$

Thus

$$\lim_{x\to 1} y=2, \text{ or, } y\to 2 \text{ as } x\to 1.$$

Example-2: Show that the following limit does not exist

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$$

Solution: The limit does not exist if it is not finite, or if it depends on a particular path $y=mx$. Consider the path . As $(x,y)\to(0,0)$, we get $x\to 0$. Therefore,

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}=\lim_{x\to 0}\frac{mx^2}{x^2(1+m^2)}=\frac{m}{1+m^2}$$

which depends on m. For different values of m, we obtain different limits. Hence the limit does not exist.

4. Differentiation

Introduction. Rate of Change. The subject of Differential Calculus which had its origin mainly in the geometrical problem of the determination of a tangent at a point of curve, has rendered possible the precise formulation of a large number of physical concepts such as *Velocity at a point*, *Acceleration at a point*, *Curvature at a point*, *Density at a point*, *Specific heat at any temperature*, etc. each of which appears as a **Local or instantaneous Rate of change** as against the **Average Rate of Change** which pertains to a finite interval of space or time and not to an instant of time and space.

The fundamental idea of **Local or instantaneous Rate of Change** pervading all these concepts underlies the analytical definition of differential co-efficient.

Derivability. Derivative. We consider a function $f(x)$ defined in any interval (a, b). Let c be any number of this interval so that $f(c)$ is the corresponding value of the function. We take $c+h$ any other number of this interval which lies to the right or left of c (*i.e.*, $c+h>$ or $<c$) according as h is positive or negative. The value of the function corresponding to it is $f(c+h)$.

Now, h, is the change in the independent variable x, and

$$f(c+h)-f(c)$$

is the corresponding change in the dependent variable $f(c)$.

The expression $[f(c+h)-f(c)]/h$, which is the ratio of these two changes, is a function of h and is not defined for $h=0$; c being a fixed number.

It is possible that the ratio tends to a limit as h tends to 0. This limit, if it exists, is called the derivative of $f(x)$ for $x=c$ and the function, then, is said to be derivable for this value.

Def. *$f(x)$ is said to be* **derivable** *at $x=c$ if*

$$\lim_{h \to 0} \frac{f(c+h)-f(c)}{h}$$

exists and the limit is called the **Derivative** *or* **Differential co-efficient** *of the function $f(x)$ for $x=c$.*

Examples:

Ex-1 *Show that x^2 is derivable for $x=1$ and obtain its derivative for $x=1$.*

Solution:

Let

$$f(x)=x^2 \text{ so that } f(1)=1^2=1.$$

To find the derivative for $x=1$, we change x from 1 to $1+h$ so that the function changes from 1 to $(1+h)^2$.

Change in the function $=(1+h)^2-1=2h+h^2$.

$$\therefore \quad \frac{f(1+h)-f(1)}{h}=\frac{2h+h^2}{h}=2+h, \text{ [as } h\neq 0]$$

which approaches 2 as h approaches 0.

Hence $f(x)$ is derivable and its derivative is 2 for $x=1$.

Ex-2 *Show that $|x|$ is not derivable for $x=0$.*

Let

$$f(x)=|x| \text{ so that } f(0)=0.$$

It will be shown that the limit of $[f(0+h)-f(0)]/h$ does not exist, when h tends to 0.

Now

$$f(h)=\begin{cases} h, & \text{if } h>0\,; \\ -h. & \text{if } h<0. \end{cases}$$

$$\therefore \quad \frac{f(0+h)-f(0)}{h}=\frac{f(h)}{h}=1 \text{ or } -1$$

according as h is positive or negative.

Hence, $[f(0+h)-f(0)]/h \to 1$ as $h \to 0$ through positive values and $\to -1$ as $h \to 0$ through negative values.

Thus $[f(0+h)-f(0)]/h$ approaches different limits when h approaches 0 through positive or negative values so that it does not tend to a limit as h tends to 0.

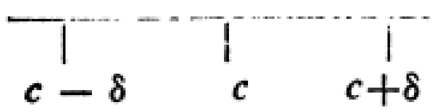
Hence $|x|$ is *not* derivable for $x=0$.

5. MAXIMUM AND MINIMUM

Maximum value of a function. Let, c, be any interior point of the interval of definition of a function $f(x)$. Then we say that $f(c)$ is a maximum value of $f(x)$, if it is the greatest of all its values for values of x lying in some neighbourhood of c. To be more definite and to avoid the vague words 'Some neighbourhood', we say that $f(c)$ is a *maximum value of the function, if there exists some interval* $(c-\delta, c+\delta)$ *around c such that*

$$f(c) > f(x)$$

for all values of x, other than c, lying in this interval.



This, again, is equivalent to saying that $f(c)$ is a maximum value of $f(x)$, if

$$f(c) > f(c+h), \text{ i.e., } f(c+h)-f(c) < 0$$

for values of h lying between $-\delta$ and δ, *i.e.*, for values of h sufficiently small in numerical value.

Minimum value of a function. $f(c)$ is said to be a minimum value of $f(x)$, if it is the least of all its values for values of x lying in some neighbourhood of c.

This is equivalent to saying that $f(c)$ *is a minimum value of* $f(x)$ *if there exists a positive* δ *such that*

$$f(c) < f(c+h), \text{ i.e., } f(c+h)-f(c) > 0$$

for values of h lying between $-\delta$ and δ, *i.e.*, for values of h sufficiently small in numerical value.

A necessary condition for extreme values. *To prove that a necessary condition for $f(c)$ to be an extreme value of $f(x)$ is that*

$$f'(c)=0.$$

Proof:

Let $f(c)$ be a maximum value of $f(x)$.

There exists an interval $(c-\delta, c+\delta)$, around c, such that, if, $c+h$ is any number, belonging to this interval, we have

$$f(c+h)<f(c),$$

Here, h may be positive or negative. Thus

$$\frac{f(c+h)-f(c)}{h}<0 \text{ if } h>0, \qquad \dots (i)$$

$$\frac{f(c+h)-f(c)}{h}>0 \text{ if } h<0. \qquad \dots (ii)$$

If h tends to 0 through positive values, we obtain from (*i*),

$$f'(c)\leqslant 0. \qquad \dots (iii)$$

If h tends to 0 through negative values, we obtain from (*ii*),

$$f'(c)\geqslant 0. \qquad \dots (iv)$$

The relations (*iii*) and (*iv*) will simultaneously be true, if and only if

$$f'(c)=0.$$

It can similarly be shown that $f'(c)=0$, if $f(c)$ is minimum value of $f(x)$.

Note 1. Geometrically interpreted, the necessary condition for extreme values obtained above states : "*Tangent to a curve at a point P where the ordinate is a maximum or minimum is parallel to x-axis.*"

Note 2. *The vanishing of $f'(c)$ is only a necessary but not sufficient condition for $f(x)$ to be an extreme value.* To see this, we consider the function $f(x)=x^3$ for $x=0$.

For value of x greater than 0, $f(x)$ is positive and is, therefore, greater than $f(0)$ which is 0 ; and for values of x less than 0, $f(x)$ is negative and is, therefore, less than $f(0)$.

Thus $f(0)$ is not an extreme value even though $f'(0)=0$.

Note 3. Stationary value. *A function $f(x)$ is said to be stationary for $x=c$ if the derivative $f'(x)$ vanishes for $x=c$, i e.,* if $f'(c)=0$; also then $f(c)$ is said to be a *stationary* or a *turning* value of $f(x)$. The term stationary arises from the fact that the *rate of change $f'(x)$ of the function $f(x)$ with respect to x is zero* for a value of x for which $f(x)$ is stationary.

Examples:

Ex.1

Find the greatest and least values of

$$3x^4-2x^3-6x^2+6x+1$$

in the interval $[0, 2]$.

Solution:

Let

$$f(x)=3x^4-2x^3-6x^2+6x+1.$$

$\therefore$

$$f'(x)=12x^3-6x^2-12x+6$$
$$=6(x-1)(x+1)(2x-1).$$

Thus

$$f'(x)=0 \text{ for } x=1, -1, \tfrac{1}{2}.$$

The value $x=-1$ does not belong to the interval $[0, 2]$ and is not, therefore, to be considered.

Now

$$f(1)=2, f(\tfrac{1}{2})=39/16.$$

Also

$$f(0)=1, f(2)=21.$$

Thus the least value is 1 and the greatest value is 21.

Ex.2

Examine the polynomial

$$10x^6-24x^5+15x^4-40x^3+108$$

for maximum and minimum values.

Solution:

Let

$$f(x)=10x^6-24x^5+15x^4-40x^3+108.$$

$$\therefore \quad f'(x)=60x^5-120x^4+60x^3-120x^2$$
$$=60x^2(x^3-2x^2+x-2)$$
$$=60x^2(x^2+1)(x-2)$$

Thus, $f'(x)=0$ for $x=0$ and $x=2$ so that we expect extreme values of $f(x)$ for $x=0$ and 2 only.

Now,

for $x<0$, $f'(x)<0$;
for $0<x<2$ $f'(x)<0$;
for $x>2$, $f'(x)>0$.

Here, $f'(x)$ does not change sign as x passes through 0 so that $f(0)$ is neither a maximum nor a minimum value.

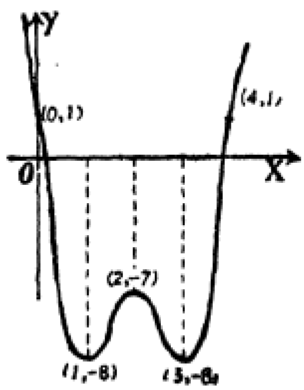
Also, since $f'(x)$ changes sign from negative to positive as x passes through 2, therefore $f(2)=-100$ is a minimum value.

Ex.3

Find the extreme values of

$$x^4-8x^3+22x^2-24x+1$$

Solution:



Let

$$y=f(x)=x^4-8x^3+22x^2-24x+1.$$

$$\therefore \quad f'(x)=4x^3-24x^2+44x-24$$
$$=4(x^3-6x^2+11x-6)$$
$$=4(x-1)(x-2)(x-3)$$

so that $\quad f'(x)=0$ for $x=1, 2, 3$.

Therefore, $f(x)$ can have extreme values for $x=1, 2, 3$ only.

Now, for $x<1$, $\quad f'(x)<0$;

for $1<x<2$, $\quad f'(x)>0$;

for $2<x<3$, $\quad f'(x)<0$;

for $x>3$, $\quad f'(x)>0$.

Since $f'(x)$ changes sign from negative to positive as x passes through 1 and 3, therefore $f(1)=-8$ and $f(3)=-8$ are the two minimum values.

Again since, $f'(x)$ changes sign from positive to negative as x passes through 2, therefore, $f(2)=-7$ is a maximum value.

Problems

Ex.1

Find the maximum and minimum values of the polynomial
$$8x^5-15x^4+10x^2.$$

Solution:

Let $\quad f(x)=8x^5-15x^4+10x^2.$

$\therefore \quad f'(x)=40x^4-60x^3+20x$
$$=20x(2x^3-3x^2+1)$$
$$=20x(x-1)^2(2x+1).$$

Hence $\quad f'(x)=0$ for $x=0, 1, -\frac{1}{2}$.

Again $\quad f''(x)=160x^3-180x^2+20$
$$=20(8x^3-9x^2+1).$$

Now, $f''(-\frac{1}{2})=-45$ which is negative so that $f(-\frac{1}{2})=\frac{21}{16}$ is a maximum value.

Again, $f''(0)=20$ which is positive so that $f(0)=0$ is a minimum value.

As $f''(1)=0$, we have to examine $f'''(1)$.

Now $\quad f'''(x)=480x^2-360x.$

$\therefore \quad f'''(1)=120$ which is not zero.

Hence $f(1)$ is neither a maximum nor a minimum value.

Ex.2

Investigate for maximum and minimum values the function
$$\sin x+\cos 2x.$$

Solution:

Let
$$y=\sin x+\cos 2x. \qquad \ldots(i)$$

$\therefore \quad dy/dx=\cos x-2\sin 2x$
$$=\cos x-4\sin x\cos x.$$

Putting $\quad dy/dx=0$, we get
$$\cos x=0 \text{ or } \sin x=\tfrac{1}{4}.$$

We consider values of x between 0 and 2π only, for the given function is periodic with period 2π.

Now $\quad \cos x = 0$ gives $\quad x = \dfrac{\pi}{2}$ and $\dfrac{3\pi}{2}$

and $\quad \sin x = \frac{1}{4}$ gives $x = \sin^{-1}\frac{1}{4}$ and $\pi - \sin^{-1}\frac{1}{4}$,

$\sin^{-1}\frac{1}{4}$ lying between 0 and $\pi/2$.

Now $\quad d^2y/dx^2 = -\sin x - 4\cos 2x$.

For $x = \pi/2$, $d^2y/dx^2 = 3$ which is positive ;

For $x = 3\pi/2$, $d^2y/dx^2 = 5$ which is positive ;.

For $x = \sin^{-1}\frac{1}{4}$ and $\pi - \sin^{-1}\frac{1}{4}$,

$d^2y/dx^2 = -\sin x - 4(1 - 2\sin^2 x) = -15/4$

which is negative.

Therefore y is a maximum for $x = \sin^{-1}\frac{1}{4}$, $\pi - \sin^{-1}\frac{1}{4}$ and is a minimum for $x = \pi/2, 3\pi/2$.

Putting these values of x in (i), we see that $\frac{9}{8}$, $\frac{9}{8}$ are the two maximum values and 0, -2 are the two minimum values.

6. GAMMA, BETA AND ERROR FUNCTIONS

Introduction:

Algebraic function $f(x)$ is obtained by the algebraic operations of addition, subtraction, multiplication, division and square rooting of x polynomial and rational functions are such functions. Transcendental functions include trigonometric functions (sine, cosine, tan) exponential, logarithmic and hyperbolic functions.

Many integrals which can not be expressed in terms of elementary functions can be evaluated in terms of beta and gamma functions.

Heat equation, wave equation and Laplace's equation with cylindrical symmetry can be solved in terms of Bessel's functions, with spherical symmetry by Legendre polynomials. We consider Fourier-Legendre series and Fourier-Bessel series. Chebyshev-polynomials which are useful in approximation theory are also presented.

6.1 GAMMA FUNCTION

Gamma function denoted by $\Gamma(p)$ is defined by the improper integral which is dependent on the parameter p,

$$\Gamma(p) = \int_0^\infty e^{-t} t^{p-1}\, dt, \quad (p > 0) \tag{1}$$

Gamma function is also known as Euler's integral of the second kind.

Integrating by parts

$$\begin{aligned}\Gamma(p+1) &= \int_0^\infty e^{-t} t^p\, dt \\ &= -e^{-t} t^p \Big|_0^\infty + p\int_0^\infty e^{-t} t^{p-1}\, dt \\ &= 0 + p\Gamma(p)\end{aligned}$$

Thus $$\Gamma(p+1) = p\Gamma(p) \tag{2}$$

By definition

$$\Gamma(1) = \int_0^\infty e^{-t}\, dt = \left.\frac{e^{-t}}{-1}\right|_0^\infty = 1 \qquad (3)$$

By the reduction formula (2),

$$\Gamma(2) = 1 \cdot \Gamma(1) = 1$$

and $\qquad \Gamma(3) = 2 \cdot \Gamma(2) = 2 \cdot 1 = 2!$

and in general when p is a positive integer n

$$\begin{aligned}\Gamma(n+1) &= n\Gamma(n) = n(n-1)\Gamma(n-1)\\ &= n(n-1)(n-2)\Gamma(n-2)\\ &= n\cdot(n-1)(n-2)\cdots 3\cdot 2\cdot 1 = n!\end{aligned}$$

Some Results of Gamma Function:

1. $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

By definition $\Gamma\left(\frac{1}{2}\right) = \int_0^\infty t^{-\frac{1}{2}} e^{-t}\, dt$, put $t = u^2$

then $\Gamma\left(\frac{1}{2}\right) = 2\int_0^\infty e^{-u^2} du$

$$\begin{aligned}\Gamma\left(\frac{1}{2}\right)\cdot\Gamma\left(\frac{1}{2}\right) &= \left[2\int_0^\infty e^{-u^2} du\right]\left[2\int_0^\infty e^{-v^2} dv\right]\\ &= 4\int_0^\infty\int_0^\infty e^{-(u^2+v^2)} du\, dv\end{aligned}$$

This double integral in the first quadrant is evaluated by changing to polar coordinates $u = r\cos\theta$, $v = r\sin\theta$, $J = r$

$$\left[\Gamma\left(\frac{1}{2}\right)\right]^2 = 4\int_{\theta=0}^{\frac{\pi}{2}}\int_{r=0}^{\infty} e^{-r^2} r\,dr\,d\theta$$

$$= 4\int_0^{\frac{\pi}{2}} -\frac{1}{2}e^{-r^2}\Big|_{r=0}^{\infty} d\theta$$

$$= 2\int_0^{\frac{\pi}{2}} d\theta = 2\cdot\theta\Big|_0^{\frac{\pi}{2}} = 2\cdot\frac{\pi}{2} = \pi.$$

Hence $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

2. $\Gamma\left(\frac{p+1}{q}\right) = qa^{(p+1)/q}\int_0^{\infty} x^p e^{-ax^q}dx$; p, q, a are positive constants

Put $y = ax^q$ then $dy = aqx^{q-1}dx$

$$\int_0^{\infty} x^p e^{-ax^q}dx = \int_0^{\infty}\left[\left(\frac{y}{a}\right)^{\frac{1}{q}}\right]^p e^{-y}\cdot\frac{1}{aqx^{q-1}}dy$$

$$= \left[qa^{(p+1)/q}\right]^{-1}\int_0^{\infty} y^{(p+1-q)/q}e^{-y}dy$$

$$= \frac{\Gamma\left(\frac{p+1}{q}\right)}{qa^{\frac{(p+1)}{q}}}.$$

6.2 BETA FUNCTION

Beta function $\beta(p, q)$ defined by

$$\beta(p,q) = \int_0^1 x^{p-1}(1-x)^{q-1}dx, (p > 0, q > 0) \quad (1)$$

is convergent for $p > 0, q > 0$. This function is also known as Euler's integral of the first kind.

Some Results of Gamma Function:

1. Symmetry: $\beta(p, q) = \beta(q, p)$

$$\beta(p,q) = \int_0^1 x^{p-1}(1-x)^{q-1}dx. \quad \text{Put } x = 1 - y$$

$$= \int_1^0 (1-y)^{p-1} \cdot y^{q-1}(-dy)$$

$$= \int_1^0 y^{q-1}(1-y)^{p-1}dy = \beta(q, p).$$

2. Beta function in terms of trigonometric functions

$$\beta(p,q) = 2\int_0^{\frac{\pi}{2}} \sin^{2p-1} x \cdot \cos^{2q-1} x \, dx$$

Putting $x = \sin^2\theta, dx = 2\sin\theta \cdot \cos\theta \; d\theta$ and $1 - x = \cos^2\theta$,

$$\beta(p,q) = \int_0^1 x^{p-1}(1-x)^{q-1}dx$$

$$= \int_0^{\frac{\pi}{2}} \sin^{2p-2}\theta \cdot \cos^{2q-2}\theta \cdot 2$$

$$\times \sin\theta \cdot \cos\theta \, d\theta$$

$$\beta(p,q) = 2\int_0^{\frac{\pi}{2}} \sin^{2p-1}\theta \cdot \cos^{2q-1}\theta \, d\theta$$

$$= 2 \cdot I_{2p-1,2q-1}$$

$$\text{or} \quad I_{p,q} = \int_0^{\frac{\pi}{2}} \sin^p\theta \, \cos^q\theta \, d\theta$$

$$= \frac{1}{2}\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right), p > -1$$

$$, q > -1$$

3. **Relation between β and Γ functions**

$$\beta(p,q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$$

Solution: By definition

$$\Gamma(p) = \int_0^\infty x^{p-1}e^{-x}\, dx,$$

Put $x = t^2, dx = 2t\,dt$

$$\Gamma(p) = \int_0^\infty t^{2p-2} \cdot e^{-t^2} \cdot 2t\, dt$$

$$= 2\int_0^\infty t^{2p-1} \cdot e^{-t^2}\, dt$$

Then $$\Gamma(p)\,\Gamma(q) = \left[2\int_0^\infty x^{2p-1}e^{-x^2}dx\right] \times \left[2\int_0^\infty y^{2q-1}e^{-y^2}dy\right]$$

Here t, x, y are dummy variables.

$$= 4\int_0^\infty \int_0^\infty x^{2p-1} \cdot y^{2q-1} \cdot e^{-(x^2+y^2)}dx\,dy.$$

Introduce polar coordinates

$$x = r\,\cos\theta, \quad y = r\,\sin\theta.$$

As x, y vary in the first quadrant (i.e., $0 < x < \infty, 0 < y < \infty$), r varies from 0 to ∞ and θ from 0 to $\frac{\pi}{2}$. Jacobian: r

$$\Gamma(p)\,\Gamma(q) = 4\int_0^{\frac{\pi}{2}} \int_0^\infty (r\cos\theta)^{2p-1} \cdot (r\sin\theta)^{2q-1} \times e^{-r^2} \cdot r\,dr\,d\theta$$

$$= 4\left[\int_0^{\frac{\pi}{2}} \sin^{2q-1}\theta \cdot \cos^{2p-1}\theta\,d\theta\right] \times \left[\int_0^\infty e^{-r^2} \cdot r^{2p+2q-1} \cdot dr\right]$$

Using result 2 above in this page and putting $r^2 = t$

$$\Gamma(p)\,\Gamma(q) = 4 \cdot \left[\frac{1}{2}\beta(p,q)\right]\left[\frac{1}{2}\int_0^\infty e^{-t} \cdot t^{p+q-1}dt\right]$$

$\Gamma(p)\,\Gamma(q) = \beta(p,q) \cdot \Gamma(p+q)$, hence the result.

4. Prove that

$$\beta(p,q) = \beta(p+1,q) + \beta(p,q+1)$$

Solution: By definition

$$\beta(p+1,q) + \beta(p,q+1)$$

$$= \int_0^1 x^p(1-x)^{q-1}dx + \int_0^1 x^{p-1}(1-x)^q \, dx$$

$$= \int_0^1 x^{p-1}(1-x)^{q-1}\left[x + (1-x)\right]dx$$

$$= \int_0^1 x^{p-1}(1-x)^{q-1}dx = \beta(p,q).$$

5. Prove that:

$$\beta(m,n) = \frac{(m-1)!\,(n-1)!}{(m+n-1)!} \text{ for } m, n \text{ positive integers}$$

Solution:

$$\beta(m,n) = \beta(m+1,n) + \beta(m,n+1)$$

Expressing in Γ functions using (6)

$$\beta(m,n) = \frac{\Gamma(m+1)\,\Gamma(n)}{\Gamma(m+n+1)} + \frac{\Gamma(m)\,\Gamma(n+1)}{\Gamma(m+n+1)}$$

Since m and n are positive integers

$$= \frac{m!(n-1)! + (m-1)!n!}{(m+n)!}$$

$$= \frac{(m-1)!(n-1)![m+n]}{(m+n)!} = \frac{(m-1)!(n-1)!}{(m+n-1)!}$$